Correspondence Fourier-transform ghost imaging

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Fourier-transform ghost imaging (FGI) enables us to realize diffraction imaging within the Fresnel region by calculating the second-order correlations of light fields. Existing FGI schemes reconstruct the Fourier frequency spectrum via correlation or sparsity constraints, which generally require object-arm detection values with relatively high signal-to-noise ratio. In this paper, we propose a correspondence FGI scheme, which reconstructs the object's Fourier spectrum by averaging partial reference patterns corresponding to object-arm detection values of high fluctuations. This conditional-averaging method is theoretically demonstrated and then explored and verified by both simulation and experiment. Moreover, an information-theory-based explanation of the simulation results is performed by studying the Fisher information of detection signals. The proposed scheme improves the sampling efficiency and is expected to expand the application of FGI by combining it with other reconstruction techniques.

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I. INTRODUCTION

Ghost imaging (GI) first emerged as an unconventional imaging technique which uses quantum-entangled photons as the source and retrieves the image via two-photon correlation [1,2]. Later, both ghost image and diffraction were demonstrated to be realized by intensity correlation using classical light sources [3–6], which below are referred to as real-space GI and Fourier-transform GI (FGI), respectively. Attendant discussions on the role of entangled and classical sources in ghost imaging conclude that though classical sources do not have as good correlation properties as entangled ones, they are also capable of achieving high-quality real-space or Fourier-transform GI [4,5,7]. This feasibility has attracted more attention to GI with classical sources since then. For real-space GI, applications have been developed in various areas, such as remote sensing [8], three-dimensional imaging [9], imaging through scattering media [10,11], and spectral imaging [12,13]. On the other hand, FGI has also found utility in obtaining a Fraunhofer diffraction pattern of a pure-phase object within both the far field [14] and the Fresnel region [15] using incoherent light source and even x rays [16].

Traditional GI uses vanilla correlation as the reconstruction algorithm, which later has been optimized by further exploiting statistics of light fields to achieve a higher reconstruction quality [17–19]. To exploit sparsity of the object's images, compressed sensing is incorporated into GI [20,21], which effectively reduce the sufficient number of samplings to achieve a considerable imaging quality. In these methods, the object-arm detection values are needed for reconstruction. On the other hand, a scheme called correspondence imaging [22] was proposed for real-space GI, which reconstructs the image by only conditionalling averaging partial reference patterns selected according to the bucket detection values. Here, the object-arm detection values are only used for selection, while comparable performance can be achieved. Hence, this conditional averaging scheme has been utilized in several studies [23,24], and even applied to photon-limit GI [25], where the required sampling time for weak-light imaging gets significant reduction.

Roughly speaking, correspondence imaging provides a scheme which selectively records signals with more information [22] and thus enhances the practical sampling efficiency for GI. However, this idea has only been demonstrated in real-space GI cases. In this paper, we propose a corresponding FGI scheme which utilizes the conditional-averaging idea in FGI. Since the imaging principle of FGI is a little different from real-space GI, the proposed scheme is demonstrated via theoretical derivation followed by simulations and experiments. Specifically, we first perform a probability-theory-based analysis of FGI, deriving a general formula for the joint probability density function (PDF) between the light intensity of a pixel on the reference detector and the object-arm detection intensity. Then, using the joint PDF, we develop the conditional-averaging method for FGI by theoretically deriving the reconstructed result, explore and investigate it via simulation, and verify its utility in experiment. Moreover, we manage to interpret the results observed in simulations by studying the information that detection signals in both arms jointly provide about the Fourier spectrum.

It is worth noting that several studies have been done to explore the probability explanation of the real-space correspondence GI [26,27], where a joint PDF of the bucket and

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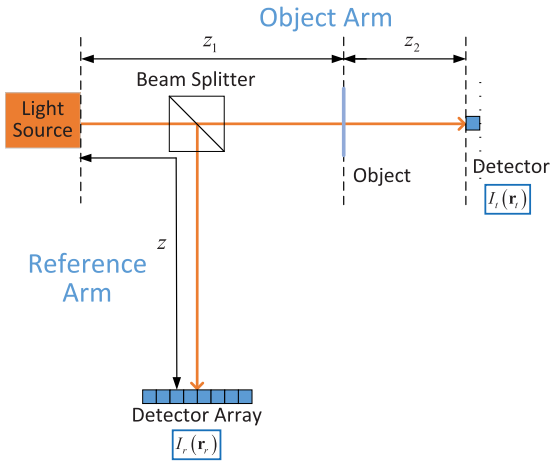


FIG. 1. Schematic diagram of FGI. Distance from the source to the reference detector array is z. In the object path, the light passes through the imaging object which is z_1 away from the source and then propagates z_2 to be recorded by a pointlike detector located at $\mathbf{r}_t$. Here, $z = z_1 + z_2$.

the reference signals is exploited. However, analyses in these studies are based on the level of light intensity and thereby do not apply to the FGI case, which essentially needs more consideration on the underlying light fields.

Remaining parts of this paper are organized as follows. In Sec. II, we perform the analysis of FGI based on probability theory by first giving a brief review of the principle of FGI in Sec. II A, then theoretically deriving the joint PDF in Sec. II B, and finally confirming the derived PDF via a FGI simulation in Sec. II C. Based on these fundamentals, in Sec. III, we develop the conditional averaging method, including deriving theoretical results of the method in Sec. III A, exploring its usage via simulation in Sec. III B, and demonstrating its utility in experiment in Sec. III C. In Sec. IV, a discussion based on information theory is presented to interpret some interesting results in the simulation. In Sec. V, we make conclusions about results shown in this paper.

II. JOINT PROBABILITY DENSITY FUNCTION OF LIGHT INTENSITIES

A. Basic principle of Fourier-transform GI

A schematic diagram of the typical FGI system is shown as Fig. 1. Light from the pseudothermal source is split into two paths. In the reference path, light directly propagates onto a spatially resolved detector, and the distance from the source to the detector is z. In the object path, the light passes through the imaging object, which is z_1 away from the source, then propagates z_2 to be recorded by a pointlike detector located at $\mathbf{r}_t$; here, $z = z_1 + z_2$. We denote $I_r(\mathbf{r}_r)$ as the light intensity distribution recorded by the spatial-resolved detector in the reference arm, where $\mathbf{r}_r$ is the spatial coordinate on the reference-arm detection plane, and $I_t(\mathbf{r}_t)$ as the light intensity detected by the point-like detector in the object arm, and then, by calculating the correlation of intensity fluctuations, the squared modulus of the Fourier transform of the object's complex transmittance distribution $t(\mathbf{r}_o)$ (with $\mathbf{r}_o$ denoting the spatial coordinate on the object plane) can be obtained [6], namely,

$$\left|\mathcal{F}\{t\}|_{\mathbf{f}=\frac{\mathbf{r}_r-\mathbf{r}_t}{\lambda z_2}}\right|^2 \propto \langle \Delta I_r(\mathbf{r}_r)\Delta I_t(\mathbf{r}_t)\rangle, \tag{1}$$

where $\Delta I_r(\mathbf{r}_r) = I_r(\mathbf{r}_r) - \langle I_r(\mathbf{r}_r)\rangle$, $\Delta I_t(\mathbf{r}_t) = I_t(\mathbf{r}_t) - \langle I_t(\mathbf{r}_t)\rangle$, $\langle\cdot\rangle$ denotes the ensemble average, and

$$\mathcal{F}\{t\}|_{\mathbf{f}=\frac{\mathbf{r}_r-\mathbf{r}_t}{\lambda z_2}} = \int t(\mathbf{r}_o)\exp\left[-i\frac{2\pi}{\lambda z_2}(\mathbf{r}_r - \mathbf{r}_t)\cdot\mathbf{r}_o\right]d^2\mathbf{r}_o. \tag{2}$$

FGI provides a method to achieve diffraction imaging without the need of far-field condition and coherent source. Based on the above principle and denotations, we perform the study based on probability theory below.

B. Derivation of the joint probability function

It can be found from Eq. (1) that Fourier components of different spatial frequencies are retrieved independently. Specifically, assuming that $\mathbf{r}_t$ is fixed, when calculating the component of a certain frequency, only the intensity values affiliated with the corresponding $\mathbf{r}_r$ are used, while others have no contribution. Hence, we focus on the joint PDF of the two intensities $I_r(\mathbf{r}_r)$ and $I_t(\mathbf{r}_t)$ in this paper. Below it can be seen that the derived result applies to arbitrary pair of $(\mathbf{r}_r, \mathbf{r}_t)$ by modifying only one parameter.

To derive the joint PDF of $I_r(\mathbf{r}_r)$ and $I_t(\mathbf{r}_t)$, a fundamental way is to deal with the corresponding light fields $E_r(\mathbf{r}_r)$ and $E_t(\mathbf{r}_t)$, since fields are the essential thing which propagates. Hence, we will first study the joint PDF of fields $E_r(\mathbf{r}_r)$ and $E_t(\mathbf{r}_t)$. To this end, an existing thing can be applied here is the joint circular complex Gaussian distribution of the fields $E(\mathbf{r})$ on the source plane, which is the key point in deriving Eq. (1) [6]. Another one to be utilized is the Fresnel diffraction formula for field propagation in both paths

$$E_r(\mathbf{r}_r) = \int d^2\mathbf{r}E(\mathbf{r})h_r(\mathbf{r}_r;\mathbf{r}), \tag{3a}$$

$$E_t(\mathbf{r}_t) = \int d^2\mathbf{r}E(\mathbf{r})h_t(\mathbf{r}_t;\mathbf{r}), \tag{3b}$$

where $h_r(\mathbf{r}_r;\mathbf{r})$, $h_t(\mathbf{r}_t;\mathbf{r})$ are impulse response functions of the reference path and the object path, respectively. With these conditions, an intuitive method is to transform the joint PDF of $E(\mathbf{r})$'s affiliated with all $\mathbf{r}$'s (essentially infinitely many) in the source plane to the joint PDF of $E_r(\mathbf{r}_r)$'s and $E_t(\mathbf{r}_t)$'s affiliated with all $\mathbf{r}_r$'s and $\mathbf{r}_t$'s (also infinitely many) with a infinite-dimensional Jacobian matrix consisting of Green's functions affiliated with different $\mathbf{r}_r$'s and $\mathbf{r}_t$'s in Eq. (2), and then reduce it to the joint PDF of two-point fields $E_r(\mathbf{r}_r)$ and $E_t(\mathbf{r}_t)$ affiliated with a fixed pair of $(\mathbf{r}_r, \mathbf{r}_t)$. However, this procedure is rather complicated. Here, we adopt a strategy to avoid that issue. Specifically, we prove that $E_r(\mathbf{r}_r)$, $E_t(\mathbf{r}_t)$ affiliated with arbitrary pair of $(\mathbf{r}_r, \mathbf{r}_t)$ are subject to a joint circular complex Gaussian distribution by confirming the following properties of their real and imaginary parts (i.e., denoting $E_r(\mathbf{r}_r) = U_r(\mathbf{r}_r) + iV_r(\mathbf{r}_r)$, $E_t(\mathbf{r}_t) = U_t(\mathbf{r}_t) + iV_t(\mathbf{r}_t)$ as commonly used in the field of statistical optics [28]):

(1) The real and imaginary parts of the field in a single location [$E_r(\mathbf{r}_r)$ or $E_t(\mathbf{r}_t)$] are subject to the same zero-mean Gaussian distributions, namely,

$$\begin{aligned} U_r(\mathbf{r}_r) &\sim \mathcal{N}\big(0, \langle U_r^2(\mathbf{r}_r)\rangle\big), \\ V_r(\mathbf{r}_r) &\sim \mathcal{N}\big(0, \langle V_r^2(\mathbf{r}_r)\rangle\big), \\ \langle U_r^2(\mathbf{r}_r)\rangle &= \langle V_r^2(\mathbf{r}_r)\rangle, \quad \forall \mathbf{r}_r; \end{aligned} \tag{4a}$$

$$\begin{aligned} U_t(\mathbf{r}_t) &\sim \mathcal{N}\big(0, \langle U_t^2(\mathbf{r}_t)\rangle\big), \\ U_t(\mathbf{r}_t) &\sim \mathcal{N}\big(0, \langle U_t^2(\mathbf{r}_t)\rangle\big), \\ \langle U_t^2(\mathbf{r}_t)\rangle &= \langle V_t^2(\mathbf{r}_t)\rangle, \quad \forall \mathbf{r}_t. \end{aligned} \tag{4b}$$

(2) Covariance (correlation) between the real and imaginary parts of the field in a single location is equal to 0, namely,

$$\langle U_r(\mathbf{r}_r)V_r(\mathbf{r}_r)\rangle = 0, \quad \forall \mathbf{r}_r; \tag{5a}$$

$$\langle U_t(\mathbf{r}_t)V_t(\mathbf{r}_t)\rangle = 0, \quad \forall \mathbf{r}_t. \tag{5b}$$

(3) Covariance (correlation) between the real parts of two fields $E_r(\mathbf{r}_r)$ and $E_t(\mathbf{r}_t)$ affiliated with arbitrary pair of $(\mathbf{r}_r, \mathbf{r}_t)$ is equal to covariance between their imaginary parts, namely,

$$\langle U_r(\mathbf{r}_r)U_t(\mathbf{r}_t)\rangle = \langle V_t(\mathbf{r}_t)V_t(\mathbf{r}_t)\rangle, \ \forall (\mathbf{r}_r, \mathbf{r}_t). \tag{6}$$

(4) Covariance between the real part of one field and the imaginary part of another field affiliated with a different location is equal to the negative covariance between the imaginary part of the first field and the real part of the second field, namely,

$$\langle U_r(\mathbf{r}_r)V_t(\mathbf{r}_t)\rangle = -\langle V_r(\mathbf{r}_r)U_t(\mathbf{r}_t)\rangle, \ \forall (\mathbf{r}_r, \mathbf{r}_t). \tag{7}$$

Details about the proofs are presented in Appendix A.

Having proved the joint circular complex Gaussian distribution, we can turn to the joint PDF of fields $E_r(\mathbf{r}_r)$ and $E_t(\mathbf{r}_t)$. Before that, we give a useful formula by recalling Eq. (1). Based on the moment theorem of circular complex Gaussian variables, one can have

$$\langle \Delta I_r(\mathbf{r}_r)\Delta I_t(\mathbf{r}_t)\rangle = |\langle E_r^*(\mathbf{r}_r)E_t(\mathbf{r}_t)\rangle|^2 \tag{8}$$

and then it can be obtained by combining with Eq. (1) that

$$|\langle E_r^*(\mathbf{r}_r)E_t(\mathbf{r}_t)\rangle|^2 \propto \left|\mathcal{F}\{t\}|_{\mathbf{f}=\frac{\mathbf{r}_r-\mathbf{r}_t}{\lambda z_2}}\right|^2. \tag{9}$$

Below for brevity, we denote the fields as E_r, E_t (and also U_r, V_r, U_t, V_t), omitting the coordinates $\mathbf{r}_r, \mathbf{r}_t$ inside the parentheses. Due to the Gaussian property, the joint PDF can be directly given based on the first- and second-order moments (means and covariances, respectively):

$$p(U_r, U_t, V_r, V_t) = \frac{1}{(2\pi)^2|C|^{1/2}} \exp\left\{-\frac{1}{2}[U_r, U_t, V_r, V_t]C^{-1}[U_r, U_t, V_r, V_t]^T\right\} \tag{10}$$

where

$$C = \begin{bmatrix} C_1 & C_2 \\ -C_2 & C_1 \end{bmatrix}; C_1 = \begin{bmatrix} \langle U_r^2\rangle & \langle U_rU_t\rangle \\ \langle U_rU_t\rangle & \langle U_t^2\rangle \end{bmatrix}; C_2 = \begin{bmatrix} 0 & \langle U_rV_t\rangle \\ -\langle U_rV_t\rangle & 0 \end{bmatrix}. \tag{11}$$

Substituting Eq. (11) into Eq. (10), we can give the joint PDF as

$$p(U_r, U_t, V_r, V_t) = \frac{\exp\left\{-\frac{\langle U_t^2\rangle(U_r^2+V_r^2)+\langle U_r^2\rangle(U_t^2+V_t^2)-2\langle U_rU_t\rangle(U_tU_t+V_rV_t)-2\langle U_rV_t\rangle(U_rV_t-V_tU_t)}{2\left(\langle U_r^2\rangle\langle U_t^2\rangle-\langle U_rU_t\rangle^2-\langle U_rV_t\rangle^2\right)}\right\}}{(2\pi)^2\left|\langle U_r^2\rangle\langle U_t^2\rangle - \langle U_rU_t\rangle^2 - \langle U_rV_t\rangle^2\right|}. \tag{12}$$

To get the joint PDF of intensities I_r and I_t, we need to get the joint PDF of amplitudes A_r and A_t (defined as below) from Eq. (12). The steps are as follows. First, substituting variables as

$$U_r = A_r\cos\theta_r, \ V_r = A_r\sin\theta_r, U_t = A_t\cos\theta_t, \ V_t = A_t\sin\theta_t, \tag{13}$$

and defining

$$\boldsymbol{\mu} = \mu e^{j\phi} = \frac{\langle E_r^*E_t\rangle}{\sqrt{\langle|E_r|^2\rangle\langle|E_t|^2\rangle}} = \frac{\langle U_rU_t\rangle}{\sqrt{\langle U_r^2\rangle\langle U_t^2\rangle}} + i\frac{\langle U_rV_t\rangle}{\sqrt{\langle U_r^2\rangle\langle U_t^2\rangle}}, \tag{14}$$

Eq. (12) can be transformed into

$$p(A_r, A_t, \theta_r, \theta_t) = \frac{A_rA_t}{(2\pi)^2\langle U_r^2\rangle\langle U_t^2\rangle(1-\mu^2)} \exp\left\{-\frac{\langle U_t^2\rangle A_r^2 + \langle U_r^2\rangle A_t^2 - 2\sqrt{\langle U_r^2\rangle\langle U_t^2\rangle}A_rA_t\mu\cos(\phi+\theta_r-\theta_t)}{2\langle U_r^2\rangle\langle U_t^2\rangle(1-\mu^2)}\right\}. \tag{15}$$

Here, two points are worth noting from the definition (14):

(1) $0 < \mu < 1$, according to the property of the complex degree of coherence;

(2) $\mu \propto |\mathcal{F}\{t\}|_{\mathbf{f}=\frac{\mathbf{r}_r-\mathbf{r}_t}{\lambda z_2}}|$, by comparing Eq. (9) and (14).

Then, by integrating (15) on θ_r, θ_t, we can obtain the joint PDF of amplitudes as

$$p(A_r, A_t) = \frac{A_rA_t}{\langle U_r^2\rangle\langle U_t^2\rangle(1-\mu^2)} \exp\left\{-\frac{\langle U_t^2\rangle A_r^2 + \langle U_r^2\rangle A_t^2}{2\langle U_r^2\rangle\langle U_t^2\rangle(1-\mu^2)}\right\} \mathrm{I}_0\left[\frac{A_rA_t\mu}{\sqrt{\langle U_r^2\rangle\langle U_t^2\rangle}(1-\mu^2)}\right], \tag{16}$$

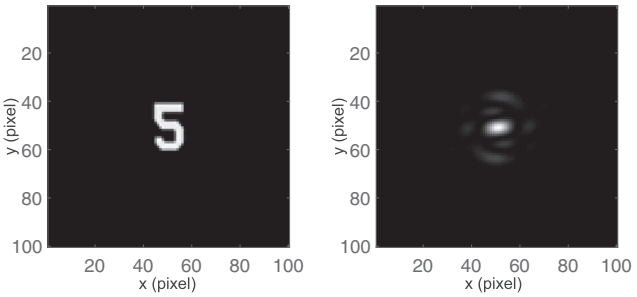


FIG. 2. (a) Image of the object (pure-amplitude) used in simulation; (b) image of the object's Fourier spectrum. Both images are 100×100 pixels, which matches the number of cells of the reference detector.

where $\mathrm{I}_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind. Again, by changing the variables as

$$I_\mathrm{r} = A_\mathrm{r}^2,\ I_\mathrm{t} = A_\mathrm{t}^2, \tag{17}$$

the joint PDF of intensities can be finally obtained as

$$p(I_\mathrm{r}, I_\mathrm{t}) = \frac{1}{\langle I_\mathrm{r}\rangle\langle I_\mathrm{t}\rangle(1-\mu^2)} \exp\left\{-\frac{\langle I_\mathrm{t}\rangle I_\mathrm{r} + \langle I_\mathrm{r}\rangle I_\mathrm{t}}{\langle I_\mathrm{r}\rangle\langle I_\mathrm{t}\rangle(1-\mu^2)}\right\} \times \mathrm{I}_0\left[\frac{2\mu\sqrt{I_\mathrm{r} I_\mathrm{t}}}{\sqrt{\langle I_\mathrm{r}\rangle\langle I_\mathrm{t}\rangle}(1-\mu^2)}\right], \tag{18}$$

where $\langle I_\mathrm{r}\rangle$, $\langle I_\mathrm{t}\rangle$ are average intensities in both paths, which are both constants independent of the coordinates $\mathbf{r}_\mathrm{r}$ and $\mathbf{r}_\mathrm{t}$. It can be seen from Eq. (18) that joint PDFs for different pairs of $(\mathbf{r}_\mathrm{r}, \mathbf{r}_\mathrm{t})$ only have difference in the parameter μ.

C. An illustration of PDF by simulation

As a supplement of the above derivation, here we present an illustration with numerically simulated data to verify the derived joint PDF as well as visualize it. The simulation is performed by calculating field propagation in the two arms of FGI shown as Fig. 1 based on Fresnel diffraction formula according to the guidance in the book [29]. Some parameters are as follows. The wavelength of the source is 532 nm, and the diameter is 2 mm, $z = 15$ cm, $z_1 = 5$ cm, $z_2 = 10$ cm. The cell size of the reference detector is $d = 15\,\mu$m, and the number of cells is 100×100. For the object, we simply choose a pure-amplitude digit 5, shown as Fig. 2(a), and its Fourier spectrum is shown as Fig. 2(b); each has a size of 100×100 pixels to match the reference detector. Here, the frequency interval in the spectrum pattern is $281.95\,\mathrm{m}^{-1}$, equal to $d/\lambda z_2$, and accordingly, the pixel size in the object image is $35.47\,\mu$m.

After obtaining the simulation data where the number of samplings equals to 50 000, we can plot the fitting joint distribution of the simulated $I_\mathrm{r}(\mathbf{r}_\mathrm{r})$ and $I_\mathrm{t}(\mathbf{r}_\mathrm{t})$ for each pair of $(\mathbf{r}_\mathrm{r}, \mathbf{r}_\mathrm{t})$ as follows. First, for both simulated I_r's and I_t's, their value ranges are both evenly divided into 100 nonoverlapping intervals, resulting in 100×100 small regions in the two-dimensional (2D) $(I_\mathrm{r}, I_\mathrm{t})$ plane. Then, we count the number of simulated samplings lying in each region and do maximum normalization among all the counts. Finally, we fit the curve with 10 000 3D points $(f, I_\mathrm{r}, I_\mathrm{t})$, where f and $(I_\mathrm{r}, I_\mathrm{t})$ are the normalized count and the central coordinates of each region, respectively. Here we present the result of $\mathbf{r}_\mathrm{t} = \mathbf{r}_\mathrm{r} = 0$, shown in Fig. 3. The left subfigure in Fig. 3(a) shows the theoretical curve of the joint PDF, and another subfigure on the right is the fitting curve of joint distribution of simulated data. Figure 3(b) is a partially enlarged version of Fig. 3(a) only containing the region where the joint probability is obviously larger than zero. In Fig. 3(b), besides the curve, we also plot a two-dimensional image at the bottom which shows the joint distribution pattern. From Fig. 3, it can be seen that the curve from the simulated data is well consistent with the theoretical one, and hence the derived joint PDF (18) is verified.

III. CONDITIONAL AVERAGE

A. Theoretical derivation

Based on the joint PDF, we can theoretically derive some results of performing a conditional averaging in FGI. Steps of derivation are as follows. Considering the case where I_t is higher than the threshold value $\beta_1\langle I_\mathrm{t}\rangle$, where β_1 is a constant, then the conditional PDF of I_r is first calculated as

$$p(I_\mathrm{r}|I_\mathrm{t} > \beta_1\langle I_\mathrm{t}\rangle) = \int_{\beta_1\langle I_\mathrm{t}\rangle}^{\infty} dI_\mathrm{t}\, p(I_\mathrm{r}, I_\mathrm{t})/P(I_\mathrm{t} > \beta_1\langle I_\mathrm{t}\rangle) = e^{\beta_1}\int_{\beta_1\langle I_\mathrm{t}\rangle}^{\infty} dI_\mathrm{t}\, p(I_\mathrm{r}, I_\mathrm{t}), \tag{19}$$

where $P(I_\mathrm{t} > \beta_1\langle I_\mathrm{t}\rangle)$ is the probability of $I_\mathrm{t} > \beta_1\langle I_\mathrm{t}\rangle$. Then, the expectation of averaging the corresponding I_r's is calculated as

$$R_+ = \int_0^{\infty} dI_\mathrm{r}\, p(I_\mathrm{r}|I_\mathrm{t} > \beta_1\langle I_\mathrm{t}\rangle) I_\mathrm{r} = \langle I_\mathrm{r}\rangle\beta_1\mu^2 + \langle I_\mathrm{r}\rangle. \tag{20}$$

From the above equation, it can be seen that the averaging is able to retrieve a positive image of the object's Fourier transform pattern with a uniform background. Next, considering a contrary case where I_t is smaller than the threshold value $\beta_2\langle I_\mathrm{t}\rangle$ (here β_2 is another constant), the conditional PDF of I_r is given as

$$p(I_\mathrm{r}|I_\mathrm{t} < \beta_2\langle I_\mathrm{t}\rangle) = \int_0^{\beta_2\langle I_\mathrm{t}\rangle} dI_\mathrm{t}\, p(I_\mathrm{r}, I_\mathrm{t})/P(I_\mathrm{t} < \beta_2\langle I_\mathrm{t}\rangle) = \frac{1}{1-e^{-\beta_2}}\int_0^{\beta_2\langle I_\mathrm{t}\rangle} dI_\mathrm{t}\, p(I_\mathrm{r}, I_\mathrm{t}) \tag{21}$$

where $P(I_\mathrm{t} < \beta_2\langle I_\mathrm{t}\rangle)$ is the probability of $I_\mathrm{t} < \beta_2\langle I_\mathrm{t}\rangle$. Then the expectation of this averaging is calculated as

$$R_- = \int_0^{\infty} dI_\mathrm{r}\, p(I_\mathrm{r}|I_\mathrm{t} < \beta_2\langle I_\mathrm{t}\rangle) I_\mathrm{r} = \langle I_\mathrm{r}\rangle - \langle I_\mathrm{r}\rangle\frac{e^{-\beta_2}\beta_2}{1-e^{-\beta_2}}\mu^2. \tag{22}$$

Here a negative image can be obtained, also with a background. Furthermore, if a subtraction is performed on the two results, one can have

$$R_+ - R_- = \frac{\beta_1 - (\beta_1 - \beta_2)e^{-\beta_2}}{1-e^{-\beta_2}}\langle I_\mathrm{r}\rangle\mu^2, \tag{23}$$

which is an image of the Fourier frequency spectrum pattern without background. Hence, theoretically, by doing

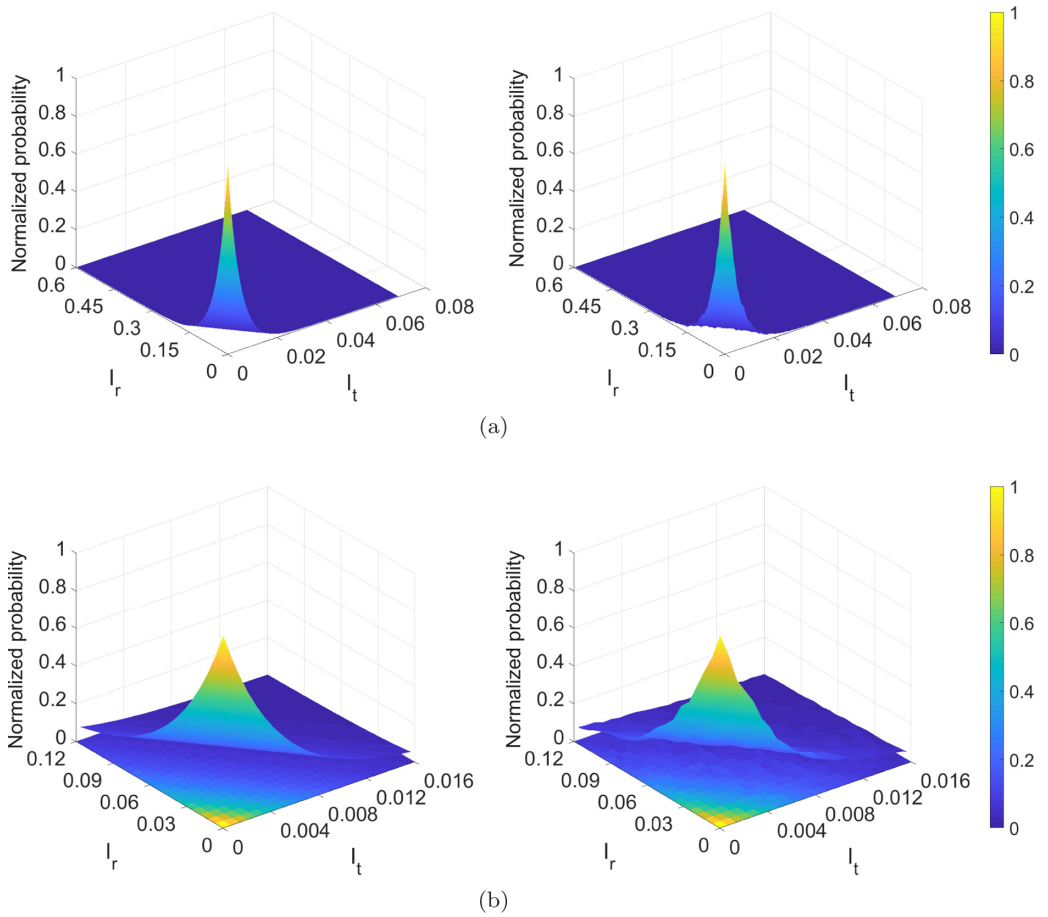


FIG. 3. Joint PDF curves of I_r, I_t affiliated with ($\mathbf{r}_r = 0, \mathbf{r}_t = 0$). (a) Comparison of the theoretical curve (shown in the left) and the fitting curve with simulated data (shown in the right); (b) partially enlarged versions of curves in Fig. (a), which only contain the region where the joint probability is obviously larger than zero, and a two-dimensional image showing the joint distribution is also plotted at the bottom in each enlarged-version figure.

conditional averaging (and subtraction) in FGI, one can also obtain the Fourier frequency spectrum.

B. Simulation results

To verify the theoretical results and explore the usage of the conditional-averaging method, we perform some calculations using the simulated data. Details of the first one are as follows. First, we divide the data into two parts according to whether I_t is larger or smaller than $\langle I_t \rangle \ln 2$, which ensures that the two parts roughly have equal sampling numbers. Then, conditional averaging is performed on the two corresponding parts of reference patterns, which retrieves positive and negative images of the Fourier spectrum, respectively. Results of positive image, negative image, and their subtraction are shown as Fig. 4. Here, the positive and negative images are shown without including the background term in (20) and (22). For comparison, an image of the Fourier spectrum is also shown here. It can be seen that the pattern of the object's Fourier spectrum can be found in these retrieved images even though with noises. Moreover, the subtraction image has a better quality compared to the other two results. These results well verify the theoretical derivation obtained in Sec. III A.

Besides the positive and negative images, in real-space correspondence imaging, it is also observed that averaging results with data whose bucket values have larger fluctuations generally have higher quality. In FGI, it can be inferred from Eqs. (20) and (22) that there should be a similar counterpart; specifically, by increasing β_1 or decreasing β_2, contrast of the positive or negative image will get enhanced. Nevertheless, averaging results under different conditions are not so clear from above theoretical analyses. Here, using the previous simulated data, we also perform a test to check what the result would be if we calculate the conditional averaging with data having different levels of fluctuations. In particular, we divide the simulated data into four parts according to the value of I_t's using the cutoff values $\langle I_t \rangle \ln \frac{4}{3}$, $\langle I_t \rangle \ln 2$, and $\langle I_t \rangle \ln 4$ (which are referred to as parts 1, 2, 3, and 4 below, respectively), so that each part has roughly equal number of data. Then, we calculate the averaging results on those four parts of reference patterns. The results are shown in Fig. 5, where Figs. 5(a), 5(b), 5(c), and 5(d) show the results of parts 1, 2, 3, and 4, respectively. It can be seen that averaging results from the reference patterns whose corresponding I_t's have larger fluctuation (parts 1 and 4) show better spectrum patterns (a negative one and a positive one). Meanwhile, it seems a little

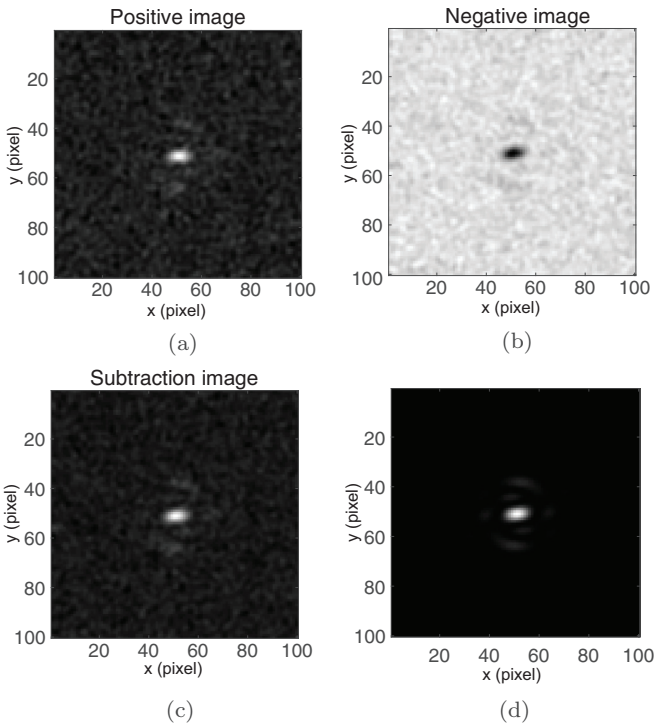


FIG. 4. Panels (a), (b), and (c) are the positive, negative, and subtraction images of the object's Fourier spectrum obtained via conditional averaging on two equally divided parts, respectively. Figure (d) shows the image of the ground-truth Fourier spectrum.

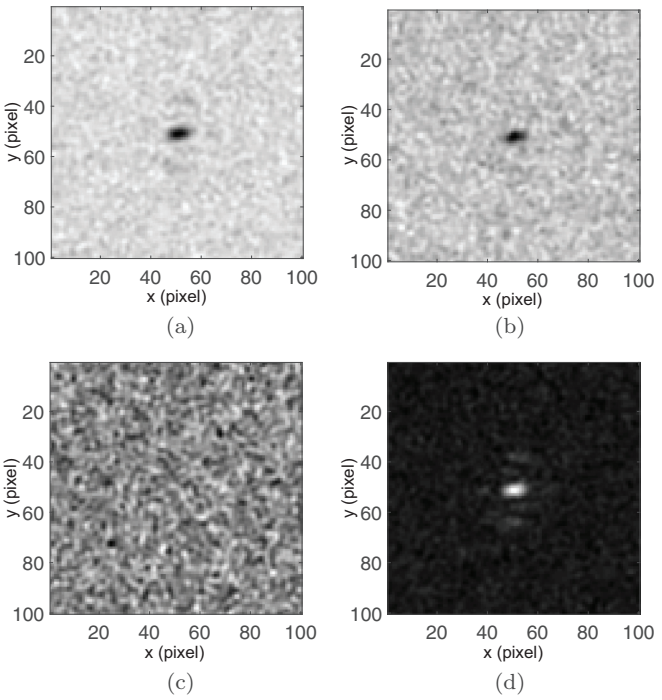


FIG. 5. Positive and negative images obtained via conditional averaging on different parts of simulated data. (a) Result from data subject to $I_t < \langle I_t \rangle \ln \frac{4}{3}$; (b) result from data subject to $\langle I_t \rangle \ln \frac{4}{3} < I_t < \langle I_t \rangle \ln 2$; (c) result from data subject to $\langle I_t \rangle \ln 2 < I_t < \langle I_t \rangle \ln 4$; and (d) result from data subject to $\langle I_t \rangle \ln 4 < I_t$.

strange that the image shown in Fig. 5(c) appears to be a random one with almost no appearance of the spectrum pattern. These results will be further explained in the discussion section.

Results of the second simulation shown above suggest that the reference patterns whose corresponding object-arm detection values have smaller fluctuations are less useful for reconstruction. Hence, by selectively sampling signals according to an appropriate threshold value, the sampling efficiency can be enhanced. To show this, we do reconstruction using the same number of samplings under the proposed conditional-averaging scheme and the vanilla correlation scheme. For the proposed scheme, the threshold values for positive and negative averaging are set as $\overline{I}_t \ln 8/7$ and $\overline{I}_t \ln 8$, where $\overline{I}_t$ is a estimated mean value using the first 100 samplings, and the final result is the subtraction of the positive and negative images. Reconstructed results of the two schemes from 15 000 samplings are shown in Fig. 6. It can be observed that, compared to the GI image, the spectrum pattern retrieved by the proposed scheme has a better quality, in both the noise level and the visibility of those weaker Fourier components. Results of this simulation reveal that the proposed scheme of selectively sampling signals with larger fluctuations is able to improve the reconstruction quality of the spectrum pattern.

C. Experimental results

To demonstrate the utility of the proposed scheme, we perform an FGI experiment with a pseudothermal source. Schematic of the experiment is shown as Fig. 1. The pseudothermal source is generated by illuminating a rotating ground glass with a 532-nm laser beam whose diameter is 5 mm. Those distance parameters in two arms are as follows: $z = 42.5\,\text{cm}$, $z_1 = 32.5\,\text{cm}$, and $z_2 = 10\,\text{cm}$. Thus, the coherence length of the light field on the detection plane is about $45.2\,\mu\text{m}$. As for the detection, we use two identical charge-coupled devices (CCDs) (Photometrics CoolSNAP cf CCD) in both arms. The cell size of the detector is $4.6\,\mu\text{m}$.

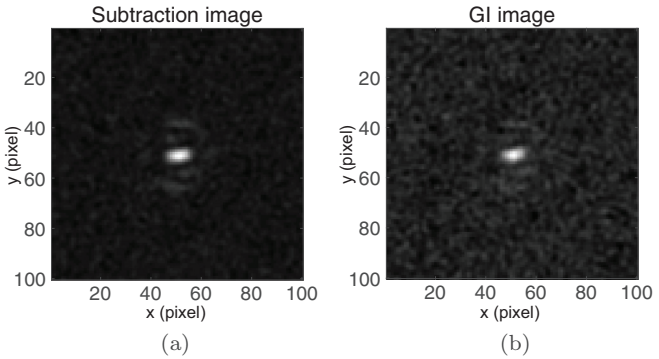


FIG. 6. Comparison of simulated results reconstructed from 15 000 samplings using the proposed scheme and vanilla correlation scheme. (a) Result of the proposed scheme; (b) result of the vanilla correlation scheme. For the proposed scheme, the threshold values for positive and negative averaging are set as $\overline{I}_t \ln 8/7$ and $\overline{I}_t \ln 8$, where $\overline{I}_t$ is a estimated mean value using the first 100 samplings, and the final result is the subtraction of the positive and negative images.

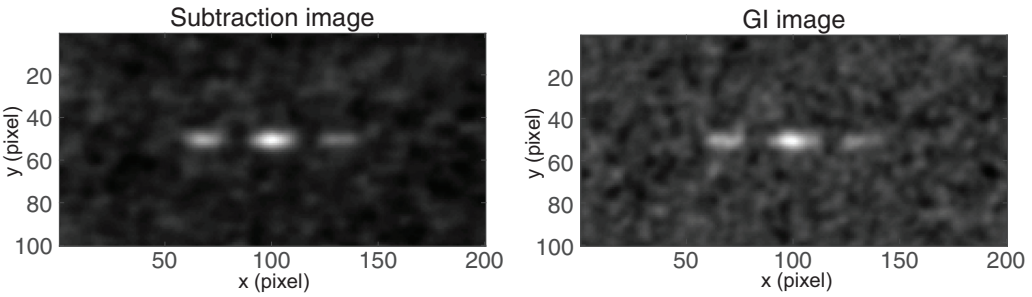


FIG. 7. Comparison of experimental results reconstructed from 15 000 samplings using the proposed scheme and vanilla correlation scheme. (a) Experimental result of the proposed scheme; (b) experimental result of the vanilla correlation scheme. The imaging object is a double-slit sample where both slits are 80 μm wide and distance between them is 160 μm. Details for performing the two schemes are the same as the last simulation.

The object is a double-slit sample, where both slits are 80 μm wide and the distance between their center is $l = 160\,\mu$m.

During the sampling process, intensity values of central 200×400 cells on the reference detector are recorded and then rescaled with a factor of 0.5, as the reference intensity patterns. For the object-arm detection, we only use the intensity value of the central four cells, and compute their average as the object-arm detection value. Hence, the equivalent pixel size on the detected pattern is 9.3 μm. In the experiment, for the proposed scheme, we also set threshold values for positive and negative averaging as $\overline{I_t} \ln 8/7$ and $\overline{I_t} \ln 8$, respectively, after estimating $\overline{I_t}$ using the first 100 samplings. Reconstructed results from 15 000 samplings using the correspondence scheme and the vanilla correlation scheme are shown in Fig. 7. It can be found that the proposed scheme produces a much better spectrum than the vanilla correlation. Specifically, the diffraction pattern obtained by the proposed scheme is much clearer and the background noise has a significantly lower level than that in the GI image. From the subtraction image, it can be observed that the spacing between the zeroth-order peak and first-order zero point is about 18 pixels (e.g., 167.4 μm), which is consistent with its theoretical value $\lambda z_2/2l = 166.3\,\mu$m. Above results show that the correspondence scheme is effective for improving the reconstruction quality of the diffraction spectrum in practical experiment.

IV. DISCUSSION

From simulation results, it is found that the quality of the reconstructed image by conditional averaging is related to the condition (i.e., the range of I_t's which correspond to reference patterns involved in averaging). Here, we try to give a interpretation of that based on information theory, since in principle, the more information, the better reconstruction. Mutual information (MI) [30] between $\gamma = \mu^2$ and (I_r, I_t), defined as

$$I(I_r, I_t; \gamma) = \iiint d\gamma dI_r dI_t p(I_r, I_t; \gamma) p(\gamma) \times \ln \frac{p(I_r, I_t; \gamma)}{\int p(I_r, I_t; \gamma) p(\gamma) d\gamma}, \tag{24}$$

with $p(I_r, I_t; \gamma)$ being the joint PDF derived above and $p(\gamma)$ denoting the PDF of γ, is a formal quantity to measure the amount of obtained information about γ by detecting (I_r, I_t). MI has been used in real-space GI to study the sampling efficiency [31] and the variation of reconstruction quality with increasing sampling number [32]. However, calculating MI is to some extent intractable here since the PDF of γ is unknown. Apart from MI, Fisher information (FI) [33] can also give a measure of the information that (I_r, I_t) provide about γ; from the perspective of statistical estimation, FI can generally indicate how well we can estimate γ from these detection values. Given the joint PDF $p(I_r, I_t; \gamma)$, the FI with respect to γ is calculated by

$$\mathcal{I}(\gamma) = E_{I_r, I_t}\left\{\left[\frac{\partial \ln p(I_r, I_t, \gamma)}{\partial \gamma}\right]^2\right\} = \iint p(I_r, I_t, \gamma)\left[\frac{\partial \ln p(I_r, I_t, \gamma)}{\partial \gamma}\right]^2 dI_r dI_t. \tag{25}$$

Compared to MI, FI is a casewise quantity depending on the specific value of γ so that one can calculate the FI corresponding to each γ value to have a more detailed interpretation. Moreover, its calculation only involves $p(I_r, I_t; \gamma)$, which has been theoretically derived. Hence, here we analyze the amount of information by calculating FI. It is also worth noting that the connection between FI and MI has been explored in the study field of neural coding [34,35], which shows that they have a close relation. Though these studies only focus on some specific types of detection models or PDFs, they provide the possibility to make further study of this in the specific GI case.

To interpret the simulation results, we manage to calculate FI's of different parts of (I_r, I_t)'s subject to different conditions. For a specific part of (I_r, I_t)'s subject to $\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle$, steps for calculating the FI are as below. First, the conditional joint PDF $p(I_r, I_t|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle)$ is calculated as

$$p(I_r, I_t|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle) = \frac{1}{e^{-\beta_1} - e^{-\beta_2}} \frac{1}{\langle I_r\rangle\langle I_t\rangle(1-\gamma)} \times \exp\left\{-\frac{\langle I_t\rangle I_r + \langle I_r\rangle I_t}{\langle I_r\rangle\langle I_t\rangle(1-\gamma)}\right\} \mathrm{I}_0\left[\frac{2\sqrt{\gamma}}{1-\gamma}\sqrt{\frac{I_r I_t}{\langle I_r\rangle\langle I_t\rangle}}\right]. \tag{26}$$

TABLE I. Fisher information of the four different parts corresponding to $I_t < \langle I_t\rangle \ln\frac{4}{3}$, $\langle I_t\rangle \ln\frac{4}{3} < I_t < \langle I_t\rangle \ln 2$, $\langle I_t\rangle \ln 2 < I_t < \langle I_t\rangle \ln 4$, and $\langle I_t\rangle \ln 4 < I_t$, respectively.

	Part			
γ	1	2	3	4
0.05	0.8205	0.3038	0.0425	2.3601
0.10	0.9027	0.3237	0.0521	2.1774
0.15	0.9979	0.3468	0.0695	2.1200

Second, the FI is calculated by

$$\mathcal{I}(\gamma|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle) = \int_{\beta_1\langle I_t\rangle}^{\beta_2\langle I_t\rangle} dI_t \int_0^\infty dI_r p(I_r, I_t|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle) \times \left[\frac{\partial \ln p(I_r, I_t|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle)}{\partial\gamma}\right]^2, \quad (27)$$

where

$$\frac{\partial \ln p(I_r, I_t|\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle)}{\partial\gamma} = \frac{1}{1-\gamma} - \left(\frac{I_r}{\langle I_r\rangle} + \frac{I_t}{\langle I_t\rangle}\right)\frac{1}{(1-\gamma)^2} + \frac{\mathrm{I}_1\left[\frac{2\sqrt{\gamma}}{(1-\gamma)}\sqrt{\frac{I_rI_t}{\langle I_r\rangle\langle I_t\rangle}}\right]}{\mathrm{I}_0\left[\frac{2\sqrt{\gamma}}{(1-\gamma)}\sqrt{\frac{I_rI_t}{\langle I_r\rangle\langle I_t\rangle}}\right]}\sqrt{\frac{I_rI_t}{\langle I_r\rangle\langle I_t\rangle}}\frac{\frac{1}{\sqrt{\gamma}} + \sqrt{\gamma}}{(1-\gamma)^2}, \quad (28)$$

and $\mathrm{I}_1(\cdot)$ is the first-order modified Bessel function of the first kind. However, there is no general analytic expression for the result of the definite integral like Eq. (27) as far as we know. Hence, below we perform numerical calculation instead.

First, we calculate the FI of the four parts (I_r, It) utilized in the second simulation. The numerical result is shown in Table I. From this, the results in Fig. 5 can be well interpreted. It can be observed that the fourth part has the highest FI, and hence Fig. 5(d) shows the best spectrum pattern among them. FI of the third part is rather small and close to zero, and thus the corresponding averaging image looks just like a random pattern with almost no information. For the first two parts, since part 1 has larger FI than part 2, the corresponding image has a lower level of noise.

It is also of interest how the FI changes with the intensity of I_t. From the above results, it is intuitively thought that the larger I_t deviates from $\langle I_t\rangle$, the more information it carries about the Fourier spectrum. Hence, we perform another numerical calculation of FI. We divide the intensity range of I_t (0—∞) into 20 nonoverlapping intervals using different pairs of (β_1, β_2), whose corresponding probabilities $P(\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle)$ are all equal. Then, we numerically calculate the FI of each part according to Eq. (27) using parameters (i.e., γ, $\langle I_r\rangle$, $\langle I_t\rangle$) obtained from simulation data. The result is shown in Fig. 8, where the FI value of each interval is plotted with the normalized I_t's mean value of the same interval. It can be seen that, as β goes up, the FI first decreases and then increases. When $\beta = 1$, the FI has the minimal value close to 0. Namely, the FI corresponding to (I_r, I_t), where I_t has greater fluctuation is larger, and when I_t is around the mean value, it contains almost no information about the imaging object. This convinces us to increase the sampling efficiency by performing a selective detection that filters out signals with low fluctuations in the object arm. It is also worth noting that, from Fig. 8, (I_r, I_t) corresponding to large β seems to have higher information than that corresponding to small β, which is not expected in advance.

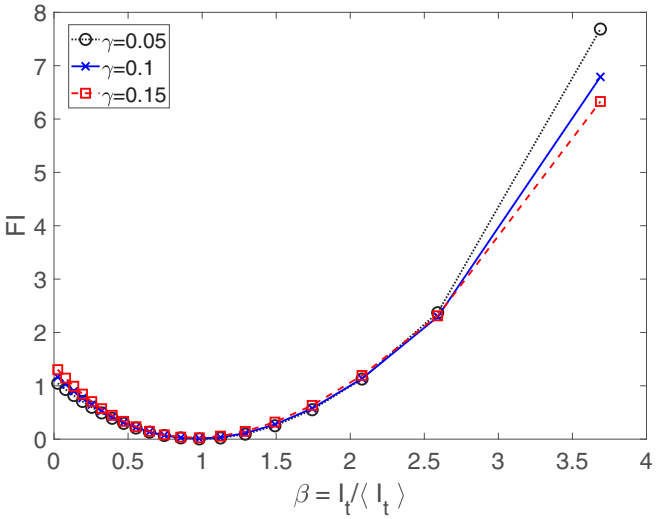


FIG. 8. The Fisher information (FI) of different parts of (I_r, I_t) as a function of I_t's relative intensity $\beta = I_t/\langle I_t\rangle$. The intensity range of I_t (0—∞) is divided into 20 nonoverlapping intervals using different pairs of threshold parameter (β_1, β_2), whose corresponding probabilities $P(\beta_1\langle I_t\rangle < I_t < \beta_2\langle I_t\rangle)$ are all equal. For each part corresponding to one interval, the FI is numerically calculated according to Eq. (27) and then plotted together with the normalized I_t's mean value of that interval.

V. CONCLUSION

In this paper, we propose and investigate the correspondence FGI scheme. We derive an general form for the joint PDF of the two intensities $I_r(\mathbf{r}_r), I_t(\mathbf{r}_t)$ in FGI. Based on the joint PDF, the conditional-averaging method for FGI is developed theoretically and verified by simulation and experiment. Both results show that, with the same number of samplings, the conditional averaging can achieve a higher quality Fourier spectrum pattern than correlation. The proposed scheme increases the sampling efficiency and is expected to expand the application of FGI by combining other reconstruction techniques. Besides, we also use Fisher information to explain the simulation results by exploring the information that object-arm detection values with different level of fluctuations carry about the Fourier spectrum, which shows that object detection signals with higher fluctuations generally provide larger information. While the derived results are based on a typical FGI system, it can be easily applied to other variants of FGI by small modifications.

ACKNOWLEDGMENTS

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APPENDIX: PROOF OF THE JOINT CIRCULAR COMPLEX GAUSSIAN DISTRIBUTION OF THE INTENSITY OF THE LIGHT FIELDS OF TWO ARMS

Given that the fields $E(\mathbf{r}) = U(\mathbf{r}) + \mathrm{i}V(\mathbf{r})$ in the source plane are subject to joint circular complex Gaussian distribution, here we prove that the fields $E_\mathrm{r}(\mathbf{r}_\mathrm{r})$, $E_\mathrm{t}(\mathbf{r}_\mathrm{t})$ shown as Eq. (3) are also subject to joint circular complex Gaussian distribution by verifying that Eqs. (4)–(7) hold. To this end, we first rewrite Eq. (3) as follows,

$$U_\mathrm{t}(\mathbf{r}_\mathrm{t}) = \int d^2\mathbf{r}\big[U(\mathbf{r})h_\mathrm{t}^\mathrm{R}(\mathbf{r}_\mathrm{t};\mathbf{r}) - V(\mathbf{r})h_\mathrm{t}^\mathrm{I}(\mathbf{r}_\mathrm{t};\mathbf{r})\big], \tag{A1a}$$

$$V_\mathrm{t}(\mathbf{r}_\mathrm{t}) = \int d^2\mathbf{r}\big[U(\mathbf{r})h_\mathrm{t}^\mathrm{I}(\mathbf{r}_\mathrm{t};\mathbf{r}) + V(\mathbf{r})h_\mathrm{t}^\mathrm{R}(\mathbf{r}_\mathrm{t};\mathbf{r})\big], \tag{A1b}$$

$$U_\mathrm{r}(\mathbf{r}_\mathrm{r}) = \int d^2\mathbf{r}\big[U(\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}) - V(\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})\big], \tag{A1c}$$

$$V_\mathrm{r}(\mathbf{r}_\mathrm{r}) = \int d^2\mathbf{r}\big[U(\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}) + V(\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})\big], \tag{A1d}$$

where

$$h_\mathrm{t}^\mathrm{R}(\mathbf{r}_\mathrm{t};\mathbf{r}) = \mathrm{Re}\{h_\mathrm{t}(\mathbf{r}_\mathrm{t};\mathbf{r})\},\ h_\mathrm{t}^\mathrm{I}(\mathbf{r}_\mathrm{t};\mathbf{r}) = \mathrm{Im}\{h_\mathrm{t}(\mathbf{r}_\mathrm{t};\mathbf{r})\},$$
$$h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}) = \mathrm{Re}\{h_\mathrm{r}(\mathbf{r}_\mathrm{r};\mathbf{r})\},\ h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}) = \mathrm{Im}\{h_\mathrm{r}(\mathbf{r}_\mathrm{r};\mathbf{r})\},$$

and list some useful equations on $E(\mathbf{r})$ resulting from the properties of circular complex Gaussian distribution,

$$\langle U(\mathbf{r})\rangle = \langle V(\mathbf{r})\rangle = 0,\ \ \forall\,\mathbf{r}; \tag{A2a}$$

$$\langle U^2(\mathbf{r})\rangle = \langle V^2(\mathbf{r})\rangle,\ \ \forall\,\mathbf{r}; \tag{A2b}$$

$$\langle U(\mathbf{r})V(\mathbf{r})\rangle = 0,\ \ \forall\,\mathbf{r}; \tag{A2c}$$

$$\langle U(\mathbf{r})U(\mathbf{r}')\rangle = \langle V(\mathbf{r})V(\mathbf{r}')\rangle,\ \ \forall\,(\mathbf{r},\mathbf{r}'); \tag{A2d}$$

$$\langle U(\mathbf{r})V(\mathbf{r}')\rangle = -\langle U(\mathbf{r}')V(\mathbf{r})\rangle,\ \ \forall\,(\mathbf{r},\mathbf{r}'). \tag{A2e}$$

Below we show the proofs.

(1) *Equation (4)*

First, according to the property of Gaussian random variables, it can be directly given that $U_\mathrm{r}(\mathbf{r}_\mathrm{r})$, $V_\mathrm{r}(\mathbf{r}_\mathrm{r})$, $U_\mathrm{t}(\mathbf{r}_\mathrm{t})$, $V_\mathrm{t}(\mathbf{r}_\mathrm{t})$ are all subject to Gaussian distribution, since they are sums of a series of Gaussian variables shown as Eqs. (A1a)–(A1d).
Then, expectation of $U_\mathrm{r}(\mathbf{r}_\mathrm{r})$ and $V_\mathrm{r}(\mathbf{r}_\mathrm{r})$ can be easily computed as

$$\langle U_\mathrm{r}(\mathbf{r}_\mathrm{r})\rangle = \int d^2\mathbf{r}\big[\langle U(\mathbf{r})\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}) - \langle V(\mathbf{r})\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})\big] = 0;\ \langle V_\mathrm{r}(\mathbf{r}_\mathrm{r})\rangle = \int d^2\mathbf{r}\big[\langle U(\mathbf{r})\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}) + \langle V(\mathbf{r})\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})\big] = 0. \tag{A3a}$$

It can be seen that there is no assumption on the specific form of $h_\mathrm{r}(\mathbf{r}_\mathrm{r};\mathbf{r})$, and thus the calculation also apply to $\langle U_\mathrm{t}(\mathbf{r}_\mathrm{t})\rangle$ and $\langle V_\mathrm{t}(\mathbf{r}_\mathrm{t})\rangle$. This analogy will also be used in the following items.
The variances can be calculated as

$$\begin{aligned}
\big\langle U_\mathrm{r}^2(\mathbf{r}_\mathrm{r})\big\rangle &= \left\langle \int d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}) - V(\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})\big]\big[U(\mathbf{r}')h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') - V(\mathbf{r}')h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}')\big]\right\rangle \\
&= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') - \langle U(\mathbf{r})V(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}') \\ -\langle V(\mathbf{r})U(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') + \langle V(\mathbf{r})V(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}')\end{bmatrix} \\
&= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') + h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') - h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}')]\end{Bmatrix};
\end{aligned} \tag{A4a}$$

$$\begin{aligned}
\big\langle V_\mathrm{r}^2(\mathbf{r}_\mathrm{r})\big\rangle &= \left\langle \int d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}) + V(\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})\big]\big[U(\mathbf{r}')h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}') + V(\mathbf{r}')h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}')\big]\right\rangle \\
&= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}') + \langle U(\mathbf{r})V(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') \\ +\langle V(\mathbf{r})U(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}') + \langle V(\mathbf{r})V(\mathbf{r}')\rangle h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}')\end{bmatrix} \\
&= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}') + h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r}') - h_\mathrm{r}^\mathrm{R}(\mathbf{r}_\mathrm{r};\mathbf{r})h_\mathrm{r}^\mathrm{I}(\mathbf{r}_\mathrm{r};\mathbf{r}')]\end{Bmatrix}.
\end{aligned} \tag{A4b}$$

By comparing Eq. (A4a) and (A4b), it can be found that

$$\big\langle U_\mathrm{r}^2(\mathbf{r}_\mathrm{r})\big\rangle = \big\langle V_\mathrm{r}^2(\mathbf{r}_\mathrm{r})\big\rangle. \tag{A5}$$

(2) *Equation (5)*

$$\begin{aligned}\langle U_{\rm r}(\mathbf{r}_{\rm r})V_{\rm r}(\mathbf{r}_{\rm r})\rangle &= \left\langle \iint d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r}) - V(\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})\big]\big[U(\mathbf{r}')h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}') + V(\mathbf{r}')h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r}')\big]\right\rangle \\ &= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}') - h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r}') + h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}')]\end{Bmatrix}. \end{aligned} \tag{A6}$$

The first term can be calculated to be equal to 0 using the permutation symmetry of integration variables $\mathbf{r}$, $\mathbf{r}'$, and the second term can be also proved to be equal to 0 by writing its conjugate form and again using the permutation symmetry of $\mathbf{r}$, $\mathbf{r}'$. Hence, we have

$$\langle U_{\rm r}(\mathbf{r}_{\rm r})V_{\rm r}(\mathbf{r}_{\rm r})\rangle = 0. \tag{A7}$$

The calculation of $\langle U_{\rm t}(\mathbf{r}_{\rm t})V_{\rm t}(\mathbf{r}_{\rm t})\rangle$ is quite similar.

(3) *Equation (6)*

$$\begin{aligned}\langle U_{\rm r}(\mathbf{r}_{\rm r})U_{\rm t}(\mathbf{r}_{\rm t})\rangle &= \left\langle \iint d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}) + V(\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})\big]\big[U(\mathbf{r}')h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + V(\mathbf{r}')h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')\big]\right\rangle \\ &= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}') + h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}') - h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}')]\end{Bmatrix}, \end{aligned} \tag{A8a}$$

$$\begin{aligned}\langle V_{\rm r}(\mathbf{r}_{\rm r})V_{\rm t}(\mathbf{r}_{\rm t})\rangle &= \left\langle \iint d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}) + V(\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})\big]\big[U(\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + V(\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')\big]\right\rangle \\ &= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')]\end{Bmatrix}. \end{aligned} \tag{A8b}$$

From Eqs. (A8a) and (A8b), we have

$$\langle U_{\rm r}(\mathbf{r}_{\rm r})U_{\rm t}(\mathbf{r}_{\rm t})\rangle = \langle V_{\rm r}(\mathbf{r}_{\rm r})V_{\rm t}(\mathbf{r}_{\rm t})\rangle. \tag{A9}$$

(4) *Equation (7)*

$$\begin{aligned}\langle U_{\rm r}(\mathbf{r}_{\rm r})V_{\rm t}(\mathbf{r}_{\rm t})\rangle &= \left\langle \iint d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r}) - V(\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})\big]\big[U(\mathbf{r}')h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + V(\mathbf{r}')h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')\big]\right\rangle \\ &= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') - h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')] \\ +\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')]\end{Bmatrix}; \end{aligned} \tag{A10a}$$

$$\begin{aligned}\langle V_{\rm r}(\mathbf{r}_{\rm r})U_{\rm t}(\mathbf{r}_{\rm t})\rangle &= \left\langle \iint d^2\mathbf{r}d^2\mathbf{r}'\big[U(\mathbf{r})h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r}) + V(\mathbf{r})h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})\big]\big[U(\mathbf{r}')h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}') - V(\mathbf{r}')h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}')\big]\right\rangle \\ &= \iint d^2\mathbf{r}d^2\mathbf{r}'\begin{Bmatrix}\langle U(\mathbf{r})U(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}') - h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}')] \\ -\langle U(\mathbf{r})V(\mathbf{r}')\rangle[h_{\rm r}^{\rm I}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm I}(\mathbf{r}_{\rm t};\mathbf{r}') + h_{\rm r}^{\rm R}(\mathbf{r}_{\rm r};\mathbf{r})h_{\rm t}^{\rm R}(\mathbf{r}_{\rm t};\mathbf{r}')]\end{Bmatrix}. \end{aligned} \tag{A10b}$$

From Eqs. (A10a) and (A10b), it can be found that

$$\langle U_{\rm r}(\mathbf{r}_{\rm r})V_{\rm r}(\mathbf{r}_{\rm r}')\rangle = -\langle V_{\rm r}(\mathbf{r}_{\rm r})U_{\rm r}(\mathbf{r}_{\rm r}')\rangle. \tag{A11}$$

From above points, it can be proved that $E_{\rm r}(\mathbf{r}_{\rm r})$ and $E_{\rm t}(\mathbf{r}_{\rm t})$ are subject to joint circular complex Gaussian distribution.

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